Resilience Academy

SDI System Installation & Customization with CRD and KoboToolbox

Theme 4: CRD Integration, Basemaps & APIs

Connecting Your SDI to the World

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Day 4 of 4 | crd.resilienceacademy.ac.tz · Mwanza + Tanga Data

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Theme 4 Overview

Goal: Use CRD as a connected, living data platform

Morning

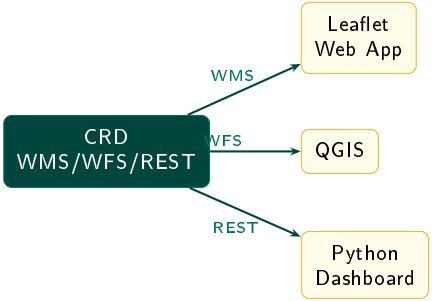
- Basemap concept: WMS/WMTS services
- Practical 4.1: Leaflet web app with CRD WMS (3h)
  - Mwanza businesses + education facilities
  - Tanga satellite imagery
  - Layer toggle and GetFeatureInfo popup

Afternoon

- Practical 4.2: CRD in QGIS + Python/GeoPandas (3h)
- Practical 4.3: CRD REST API dashboard (3h)
- Wrap-up, feedback, certificates

CRD as a Basemap: The Uber/Bolt Analogy

- Apps like **Uber and Bolt** display a background map using **WMS tile services**
- Your CRD exposes **every layer** as a WMS/WMTS endpoint — automatically
- Any WMS-compatible application can use your CRD as a basemap:
  - Leaflet (JavaScript)
  - OpenLayers
  - QGIS
  - ArcGIS
  - Mobile apps



Key Insight

Upload a layer to CRD once → instant WMS, WFS, and REST API access for all connected applications.

Leaflet Web App with CRD Basemap

HTML/JS: Mwanza & Tanga layers from CRD

```
var map = L.map('map').setView([-4.5, 35.5], 6);

// Background: OpenStreetMap
L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png').addTo(map);

var CRD = 'https://crd.resilienceacademy.ac.tz/geoserver/ows';

// CRD WMS layer -- Mwanza businesses
var mwanzaBiz = L.tileLayer.wms(CRD, {
  layers: 'geonode:mwanza_businesses',
  format:'image/png', transparent:true, opacity:0.85
}).addTo(map);

// CRD WMS -- Tanga satellite imagery
var tangaImg = L.tileLayer.wms(CRD, {
  layers: 'geonode:Tanga_imagery_reduced',
  format:'image/png', transparent:true
});
```

CRD REST API + GeoPandas Dashboard

CRD REST API

```python
import requests

CRD="https://crd.resilienceacademy.ac.tz"

# List all Mwanza datasets
r=requests.get(
  f"{CRD}/api/v2/datasets/?search=mwanza")
data=r.json()
print(f"Mwanza datasets: {data['total']}")
for ds in data['resources']:
    print(f" {ds['title']}")

# List all Tanga datasets
r2=requests.get(
  f"{CRD}/api/v2/datasets/?search=tanga")
print(f"Tanga datasets: {r2.json()['total']}")
```

GeoPandas WFS Dashboard

```python
import geopandas as gpd
import matplotlib.pyplot as plt

# Load Mwanza businesses via WFS
url=(f"{CRD}/geoserver/ows?service=WFS"
  "&version=2.0.0&request=GetFeature"
  "&typeName=geonode:mwanza_businesses"
  "&outputFormat=application/json")
gdf=gpd.read_file(url)
print(f"Loaded {len(gdf)} features")

# Map + chart
fig,ax=plt.subplots()
gdf.plot(ax=ax,color='#007064',markersize=4)
ax.set_title('Mwanza Businesses -- CRD')
plt.savefig('mwanza_dashboard.png',dpi=150)
```

4-Day Training: What You Built

Completed

- ✓ Full DevOps environment: Docker, Git, Linux
- ✓ CRD (GeoNode) installed with Docker
- ✓ CRD installed on Ubuntu bare metal
- ✓ CRD customised with branding and settings
- ✓ Tanga & Mwanza data loaded in CRD
- ✓ KoboToolbox self-hosted and configured
- ✓ Tanga flood survey form deployed
- ✓ Survey data integrated into CRD
- ✓ CRD serving as live WMS basemap
- ✓ CRD REST and WFS consumed in Python

Next Steps

- Deploy CRD + KoboToolbox on cloud VPS with real domain
- Contribute Tanga/Mwanza data to live CRD
- Explore advanced customisation at `docs.geonode.org`
- Build dashboards for your institution's data

Resources

docs.geonode.org · kobotoolbox.org/documentation
crd.resilienceacademy.ac.tz/developer/

Asante Sana!

Thank You

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